

Runtime verification of the audit findings

Huchu schools vertical audit · Runtime verification of the audit findings · reference/runtime-verification.md · rendered 2026-09-14

The first pass of this audit was a static read of the code. This second pass ran the product against a seeded local database and checked the main findings by driving it as each persona. Where the two passes disagree, the runtime result wins and the persona documents have been corrected.

Environment

Item	Value
Commit	<code>main</code> at <code>01de8a3</code> , audit branch <code>claude/school-system-audit-4u9gdn</code>
Database	PostgreSQL 16, local, built from the migration history with <code>pnpm e2e:bootstrap</code> (all 65 migrations applied cleanly, feature catalogue synced: 122 features, 21 bundles, 6 tiers)
Tenant	St Marys High School (<code>stmarys</code>), created with <code>scripts/seed-staging-tenant.ts</code> and populated with <code>scripts/seed-school-demo.ts --reset</code>
Seeded data	2026 year with 3 terms (Term 3 current), 6 classes, 13 subjects, 8 teachers, 78 class-subject assignments, 120 pupils (one suspended, 40 boarders), 119 guardians (one pupil with none), 90 attendance sessions with 1,800 marks, 156 assessments with 3,100 scores (one unmarked), 6 fee structures, 120 invoices (45 paid, 20 part-paid, 55 issued, 8 overdue), document templates seeded
Accounts	head (SUPERADMIN), three HODs and five TEACHER users linked to teacher profiles, <code>student@</code> and <code>parent@</code> portal users from the seed; <code>bursar@</code> , <code>registrar@</code> and <code>warden@</code> added for this pass with the seed password
Server	<code>pnpm dev:e2e</code> on port 300 with <code>.env.e2e</code> , hosts <code>stmarys.apps.pagka.local</code> and the three portal prefixes resolved locally; portal hosts nominated with the preview-host cookie, as the e2e suite does
Driver	Playwright with the pre-installed Chromium, service workers blocked (the repository's own portal specs do the same), viewports 1440×900, 1024×768, 390×844 and 1280×800
Unit and integration tests	<code>vitest run lib/schools app/api/v2/schools lib/documents/schools-sources.test.ts lib/id-generator-school-numbering.test.ts</code> against a local <code>huchu_test</code> database

Not exercised: ZIMRA fiscalisation (no device configured), email or SMS delivery (none exists), the Vercel build.

Automated tests

Run	Result
Schools-only suite: 34 files	743 passed, 0 failed
Whole repository suite: 197 files	196 passed; 2 tests failed in <code>lib/inventory/shelf-price-integrity.test.ts</code> (retail, unrelated to schools)

The schools domain tests pass in full. That supports the first pass's statement that the money core and the domain rules are well covered, and it does not contradict any finding, because the findings are about routes and screens the suite does not exercise (results transitions, receipt void against fiscal state, draft-invoice allocation, guardian-link PATCH, portal scoping of term marks, and every screen).

Browser checks

Each row is one check from the verification script. Verdicts: CONFIRMED means the runtime behaved as the static finding said; PARTIAL means part of the finding held; REFUTED means it did not hold and the persona document has been corrected; INCONCLUSIVE means the seeded data could not settle it; INFO is an observation with no pass or fail.

Totals: 34 confirmed, 1 partial, 2 refuted, 2 inconclusive, 2 not seen, 1 informational, across 42 checks.

Surface	Check	Finding under test	Expected	Observed	Verdict
admin	A-teachers-clip	Teachers table actions column clipped	last header right edge > viewport	The last table header was not measurable on this run; the clipped Actions column was observed in the older screenshot set only.	NOT SEEN
bursar	B-ledger-phone	Ledger on a phone shows no invoice on the first screen	first row below 844px	first row top=973px	CONFIRMED
bursar	B-clip	Row verbs clipped off the right edge on the class fees page	buttons with right edge beyond the viewport	buttons=49; offRight=28; ["Take payment", "Write off", "Take payment"]	CONFIRMED
bursar	B-nav	Sidebar is not persona-filtered: bursar sees results, boarding, attendance	foreign bands visible	foreign items= ["Scheme of work", "Homework", "Result sheets"]	CONFIRMED
bursar	B-remind-button	'Send reminders' enabled for BURSAR on the finance overview	enabled	count=1; enabled=true	CONFIRMED
bursar	B-notices	POST /notices refused for BURSAR (reports:create)	403 'cannot create reports'	status=403 {"error": "Your role cannot create reports"}	CONFIRMED
bursar	B-messages	POST /messages refused for BURSAR	403	status=403 {"error": "Your role cannot create reports"}	CONFIRMED

Surface	Check	Finding under test	Expected	Observed	Verdict
bursar	B-guardian-link	PATCH /guardian-links/[id] has no persona check (BURSAR can change consent)	200	status=200 then 200 {"id":"a265c06b-4626-41ae-8ded-aa9f4978e607","relationship":"Grandmother","isPrimary":true,"canReceiveFinancials":true,"	CONFIRMED
bursar	B-draft-receipt	A receipt saved with postNow:false is DRAFT and cannot be posted, allocated or voided	create 2xx DRAFT; post 404/405; allocate 4xx; void 4xx	create=201 receipt SFR-0001 saved as DRAFT (confirmed in the database); POST .../post=404 (no such route); .../allocate=400 (refuses a non-POSTED receipt); .../void=400 (refuses a non-POSTED receipt). The draft can never be posted, allocated or voided.	CONFIRMED
bursar	B-waiver	One bursar can create a waiver directly as APPROVED/APPLIED	2xx with status APPROVED/APPLIED	create=201; database row status=APPLIED, approvedById set and equal to createdById, appliedAt set. One bursar created, approved and applied the waiver in a single call.	CONFIRMED
bursar	B-import	Opening-balance import refused for BURSAR (needs students:create too)	403	status=403 {"error":"Your role cannot create students"}	CONFIRMED
bursar	B-receipts-primaries	Two 'Record receipt' primaries on the receipts view	>=2	count=1 on this run because a receipt already existed; the second primary belongs to the empty state (run 1 screenshot of the empty receipts view shows two "Record receipt" buttons).	PARTIAL
bursar	H-ageing	Ageing buckets differ between finance overview and arrears report	different bucket labels/values	On the seeded data the finance overview (Over 90 days \$ 3,920.00) and the arrears report (chip "90+ days 3,920", chart bar in 90+) agree. The three surfaces still compute ageing in three code paths with different bucket labels; the disagreement in the older sc	REFUTED

Surface	Check	Finding under test	Expected	Observed	Verdict
head	H-nav	Shipped sidebar shows top-level 'Whole school', a 'Scheme of work' item, and no master-data entries	Whole school present; Scheme of work present; years/classes absent	Whole school=true; Scheme of work=true; master-data links=false; items=117	CONFIRMED
head	H-overview	Overview renders 0% ratios and repeated Remind buttons	0% ratios and per-row Remind	'0%' occurrences=2; Remind buttons=6	CONFIRMED
head	H-scheme	'Scheme of work' redirects office users into the teacher portal and dead-ends	URL under /portal/teacher and 'not linked' copy	url=/portal/teacher/syllabus; the head lands in the teacher shell as "Teacher · No classes this term" with three empty pickers; the copy is not the "not linked" sentence the e2e sweep asserts, but the screen is a dead end for the head.	CONFIRMED
head	H-rationale-homework	Rationale prose rendered on /schools/homework	prose present	match=true	CONFIRMED
head	H-rationale-goals	Rationale prose rendered on /schools/goals	prose present	match=true	CONFIRMED
head	H-rationale-arrears	Rationale prose rendered on /schools/finance/arrears	prose present	match=true	CONFIRMED
head	H-rationale-notices	Rationale prose rendered on /schools/notices	prose present	match=true	CONFIRMED
head	H-rationale-meetings	Rationale prose rendered on /schools/meetings	prose present	match=true	CONFIRMED
head	H-results-phone	Results page at 390px: table header not on first screen	header below 844px	table header top=852px	CONFIRMED
head	H-attendance	Attendance oversight has two search boxes and all registers DRAFT	2 searches	search inputs=2; mentions 'not submitted'=true	CONFIRMED
head	H-notice	Head can send a notice (in-app)	201/200	status=201 {"id":"346a4fd5-2ca2-417c-92cc-bf13c4544332","recipients":1,"withoutAccount":117}	CONFIRMED
head	H-rollup	Roll up term marks into a DRAFT sheet for the student's class	200	status=201 {"sheetId":"e590cf18-33e0-4a4a-824e-8365c433b6c8","linesWritten":247,"skipped":13,"scheme":{"id":"","name":"Standard scheme"}}}	CONFIRMED

Surface	Check	Finding under test	Expected	Observed	Verdict
head	H-record	Student record fees tab deep link shape	/schools/finance?invoice=	No finance link found on the student record's default tab; the Fees tab was not opened by the script. Source: student-record-page.tsx:635.	INCONCLUSIVE
parent	P-download	Document download fails for PARENT (render route role check)	403 'Your role cannot view fees'	buttons=2; render status=403 {"error": "Your role cannot view fees"}	CONFIRMED
parent	P-attendance	'not yet submitted' repeated on every day row	>= 5 occurrences	occurrences=17	CONFIRMED
parent	P-wide	No navigation above 900px (bottom tabs hidden, no side nav)	0 visible nav links	{"visibleNavs":0,"visibleNavLinks":0}	CONFIRMED
parent	P-notice	Tapping a notice does not open a detail; only the row summary is shown, the full body is not	URL unchanged, last sentence of the body absent	The list row renders the full notice body including its last sentence; tapping it marks it read and does not open a detail view (URL unchanged). Detail screen, attachments, RSVP and replies are absent, but "the body is never shown" was wrong.	REFUTED
parent	P-home	Home renders; lead figure	-	has '\$ 0.00'=true; 'Marks'=true	INFO
student	S-leak	Unpublished marks surface via /me/subjects currentMark and goals while /me/results shows only PUBLISHED	subjects with currentMark > 0 while published lines = 0	subjects=200 withCurrentMark=13/13; results=200 publishedLines=0; goals=200 withCurrentMark=0	CONFIRMED
student	S-settings	'Change your password' link leaves the portal and is bounced back; explanatory alert present	landed under /portal/student; alert true	link=1; landed=/portal/student; alert=true	CONFIRMED
student	S-timetable	Day picker includes 'Sat' while the grid is Mon to Fri	Sat present	"Sat" not found in the rendered day picker on the seeded timetable; the DAYS constant in source (student-timetable-screen.tsx:30) includes Saturday.	NOT SEEN
teacher	T-today	Teacher portal loads for a TEACHER with a profile	loads	linked=true	CONFIRMED

Surface	Check	Finding under test	Expected	Observed	Verdict
teacher	T-register	Register save leaves the session DRAFT; no Submit control in the portal	save 2xx status DRAFT; submit buttons 0	register=200 keys=classSubject,on Date,session,rows,cou nts roll=20; save=200 status=DRAFT { "success":true,"data" :{"resource":"portal- teacher- attendance","compan yId":"291de2de-0eba- 471f-9575-; submitButton=0	CONFIRMED
teacher	T-meetings	POST /meetings refused for TEACHER (students:edit)	403	status=403 { "error": "Your role cannot edit students" }	CONFIRMED
teacher	T-notices	POST /notices ('tell the family') refused for TEACHER	403	status=403 { "error": "Your role cannot create reports" }	CONFIRMED
teacher	T-scope	GET term-marks for a class the teacher does not teach is permitted (no ownership check)	200	term-marks=200; assessments list=200	CONFIRMED
teacher	T-messages	No way to start a thread or broadcast from the teacher portal	0 start/broadcast controls	controls=0; copy='parents start them'=true	CONFIRMED
teacher	T-settings	Settings renders rows stamped 'Not yet available'	>=4	4 rows stamped "Not yet available" on the default panel. The stale "Parent messaging is not built in this portal yet" sentence sits on the Mark publishing panel (source line 283), not the default panel, so it was not in the rendered text.	CONFIRMED
teacher	T-bell	Bell badge carries papers-to-mark count rather than unread messages	bell text equals papers count	This teacher has no papers to mark on the seeded data, so the badge is empty; the code path (badge = papersToMark) could not be exercised. Source: teacher-portal-shell.tsx:271-277.	INCONCLUSIVE

Corrections made to the persona documents after this pass

1. **Parent portal, notices.** The static pass said the notices API sends only a title and summary so a notice's body is never readable. At runtime the list row renders the full body, including the last sentence of a 400-character notice. What holds: tapping a row only marks it read, there is no detail screen, and attachments, RSVP and threaded replies are absent. `parent-portal/ui-ux.md` and `parent-portal/workflows.md` corrected.

2. **Bursar, ageing.** The static pass said three ageing computations "give two different answers" for the same \$3,920. On the seeded tenant the finance overview and the arrears report agree (both place the \$3,920 in the over-90 bucket). Three separately coded bucket sets with different labels remain a maintenance risk, and the disagreement in the older screenshot set was real for that data, but it is not a live defect on this build. `bursar/workflows.md`, `bursar/ui-ux.md` and the README corrected.
3. **Head, attendance oversight.** The static pass said the attendance page had no "remind all" action. It has one ("Send all six a reminder" in the banner). The six per-row Remind buttons with no bulk action are on the overview page only. `head/ui-ux.md` corrected.
4. **Head, "Scheme of work" dead end.** Confirmed, with a nuance: the head does not see the "not linked to a teacher profile" sentence the e2e sweep asserts; they land in the teacher shell as "Teacher · No classes this term" with three empty pickers. Same outcome, different copy. `head/workflows.md` wording adjusted.
5. **Admin, teachers table.** The clipped Actions column seen in the older screenshot set was not reproduced at 1440px on the seeded tenant; the eight identical "Find the employee" buttons were. `admin/ui-ux.md` adjusted.
6. **Student portal, timetable day picker.** The Saturday chip in the `DAYS` constant did not render on the seeded timetable; left as a source observation only. `student-portal/ui-ux.md` adjusted.
7. **Bursar, receipts view.** The two "Record receipt" primaries appear only while the receipts view is empty; once a receipt exists there is one. `bursar/ui-ux.md` adjusted.
8. **Portal host routing (new observation).** On this local setup the real portal hosts (`staff.`, `parents.`, `students.` prefixes) serve the tenant workspace sign-in form at `/login` instead of the portal form; the internal paths `/portal/<portal>/login` on the tenant host serve the portal forms. The repository's own e2e fixtures record the same behaviour under host nomination and drive portals by internal path. Whether the prefix rewrite works in production is untested by the suite and was not settled here; it is added to `admin/workflows.md` as an open item.

Screenshots

The screenshots captured during this pass are kept under `docs/audits/schools/reference/runtime-shot` `s/`. They supersede the older sets under `docs/screenshots/schools/` where the two differ, because they were taken on the audited commit with the seeded tenant.

- `H-rationale-arrears.png`
- `H-rationale-goals.png`
- `H-rationale-homework.png`
- `H-rationale-meetings.png`
- `H-rationale-noticees.png`
- `admin-classes-1440.png`
- `admin-teachers-1440.png`
- `admin-timetable-1440.png`
- `bursar-arrears-1440.png`
- `bursar-class-fees-1440.png`

- bursar-class-fees-390.png
- bursar-finance-overview-1440.png
- bursar-finance-overview-390.png
- bursar-ledger-invoices-1440.png
- bursar-ledger-invoices-390.png
- bursar-nav-1440.png
- bursar-receipts-1440.png
- head-admissions-1440.png
- head-attendance-1440.png
- head-attendance-390.png
- head-moderation-1440.png
- head-overview-1440.png
- head-overview-390.png
- head-results-390.png
- head-scheme-redirect.png
- head-student-record-1440.png
- parent-attendance-390.png
- parent-fees-390.png
- parent-home-1280.png
- parent-home-390.png
- parent-marks-390.png
- parent-messages-390.png
- parent-notices-390.png
- parent-profile-390.png
- student-goals-390.png
- student-home-390.png
- student-homework-390.png
- student-marks-390.png
- student-settings-390.png
- student-timetable-390.png
- teacher-marks-1024.png
- teacher-marks-book-1024.png
- teacher-messages-1024.png
- teacher-register-1024.png
- teacher-register-390.png
- teacher-reports-1024.png
- teacher-settings-1024.png
- teacher-today-1024.png